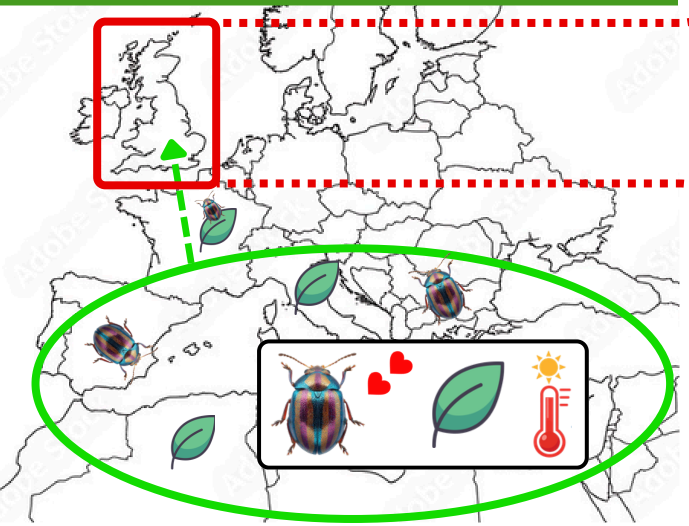
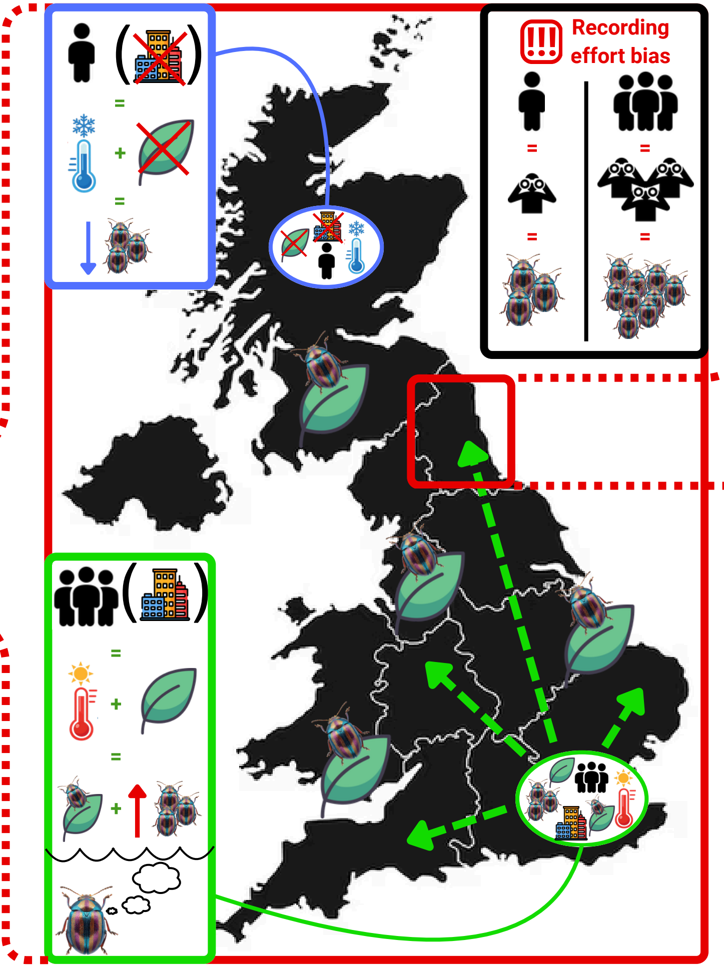
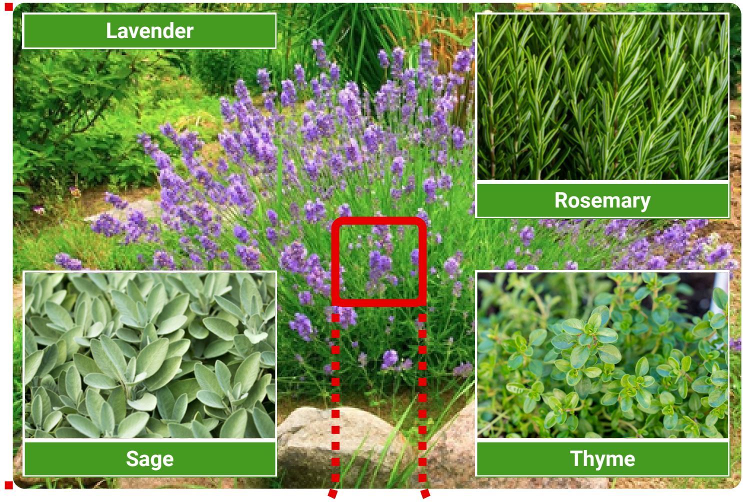
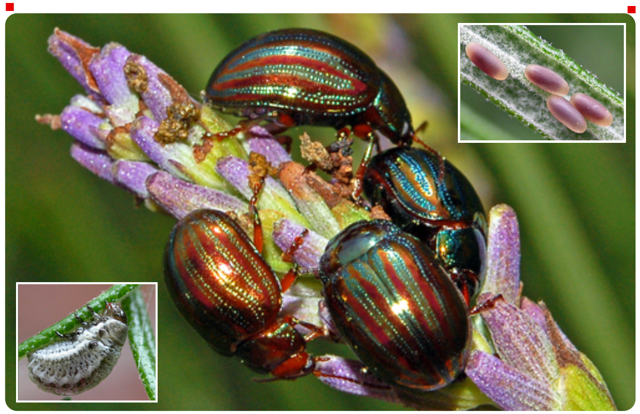
# QUANTIFYING RECORDER EFFORT, CLIMATE BASELINES, AND URBAN INFRASTRUCTURE AS DRIVERS OF THE ROSEMARY BEETLE (*CHRYSOLINA AMERICANA*) INVASION IN THE UK





## DATASETS

| BEETLE SPECIES | Target (*Chrysolina americana*) | Calibration (*Pyrochroa coccinea*) |
|---|---|---|
| Occurence (raw count) | | |
| Presence (at least 1 count in a ward) | | |
| Abundance (counts in a ward) | | |

Human population density (per ward)

(Proxy of) level of urbanisation

(Proxy of) plant habitat availability

Temperature grid (average values 1981-2000)

Additional pictograms

Density over time

Jump dispersal

Time

Observer

## METHODS

- R and ArcGIS: data analysis & spatial modelling
- Generalised linear (mixed) models account for the recorder bias using the calibration species

- In each ward, calculation of density coefficients (DC), using the ratio: ÷
- Heatmap of broad concentration then DC map are generated

H1

- Model: binomial GLM
- Aim: determine drivers for odds of establishment (presence vs absence) in a ward
- Variables:

H2

- Model: negative binomial GLMM
- Aim: determine drivers for shift in abundance in a ward
- Variables:
- simulated for =/=

H3

- Climatic increment simulations of +1.5°C, +2.5°C, and +4.0°C over historical baseline
- Predicted abundance maps are generated to highlight areas impacted by temperature shift

H4

## RESULTS

### Recording effort bias

- & maps heavily correlate
- DC low in urban hubs, confirming disproportionate observations there

### Presence

w/ +

### Abundance

w/ + +

w/ +

### Climate sensitivity analysis

- Global ↑ in predicted abundance, especially in the South & in cities
- Formation of corridors in the rural areas facilitating dispersion

### Interaction of human pop. vs time

- Annual impact of pop. density ↓
- Approaching carrying capacity in cities → density dependence, and ↑ growth in suburban/rural areas